

# SIR/SAFIA

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## 1. Introduction

## 2. System Identification and Regression (SIR)

A slightly more detailed flow of the information within SIR is demonstrated as in Figure 2. The raw observables  $u$  together with specific terms (for example,  $\log(1 + u)$ ) coming from domain knowledge of user

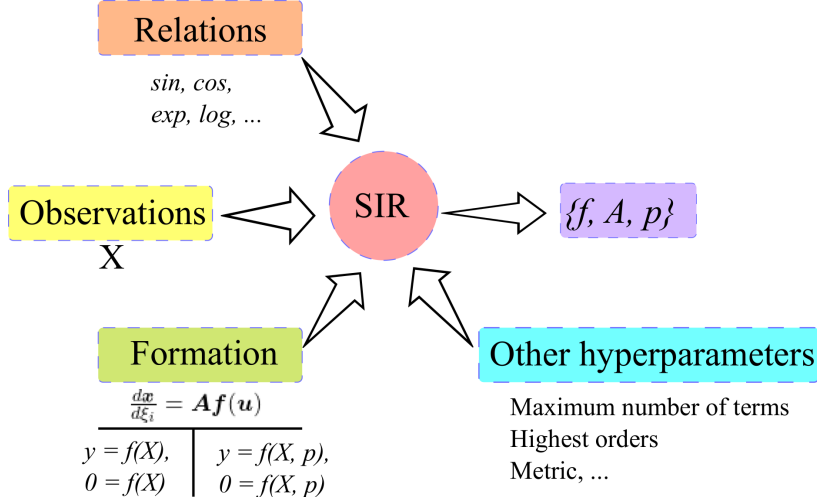


Figure 1: The Rough Input/Output flow of the proposed SIR package.  $f$ : possible mathematical expressions,  $A$ : coefficients in the expression that fits the input data,  $p$ : statistics of the possible random variables in the expression.

or suggested from other tools such as a genetic programming based symbolic regression tool are fed into the generator for generating admissible terms. Optimizer will then take on the task to produce optimized expressions. These expressions can be further tuned by post-processing unit trading accuracy for simplicity of the expression. For example, one can select  $-0.3u^2 + 1.1u \log(1 + u)$  from  $0.0005u - 0.3u^2 + 1.1u \log(1 + u) + 0.0001$  by thresholding the coefficients. The resulting terms  $u^2, u \log(1 + u)$  can be fed back as the input for a second round fit or repeated for multiple times with different specifications to generate more accurate or more complicated expressions until satisfaction.

Like the objectives for most symbolic regression tools, the general goal is to find some kind of explicit mathematical formulations/rules  $\mathbf{f}$ , so that the observed data  $\mathbf{x} = \mathbf{x}(\xi_1, \xi_2, \dots, \xi_n)$  follows it for  $n$  independent variables  $\xi_i$ . An ideal solution  $\mathbf{f} = \mathbf{f}(\mathbf{u})$  should at least satisfy the following two criterion:

- $\mathbf{x}$  should follow  $\mathbf{f}$  well enough in terms of some measure.
- $\mathbf{f} = \mathbf{f}(\mathbf{u})$  should be parsimonious, i.e. as simple a form as possible while also maintaining physical interpretability.

In many application scenarios the second may matter more than the first, due to the strong theoretical preference for *physics-based first principle models*. However, the derived set  $\mathbf{u}$  can in practice be any arbitrary collection of forms that are amenable to appropriately encode the application behavior of interest. Generally the set  $\mathbf{u}$  is derived from either functions of the  $\xi_i$ 's or  $x_i$ 's, where  $\mathbf{x}$  is a state vector of observed data,  $\mathbf{x} = \{x_1, \dots, x_m\}$  — note that all bold variables are state vectors. As examples of admissible  $\mathbf{u}$ 's, consider:

- $\mathbf{u} = \{\mathbf{x}, D\mathbf{x}, \dots, D^i\mathbf{x}, \dots, D^n\mathbf{x}, g(\boldsymbol{\xi})\}$
- $\mathbf{u} = \{\partial_{\xi_1} x_1, \partial_{\xi_2}(x_1 x_2), \partial_{\xi_1}(x_1 x_2), \partial_{\xi_2}(x_1 x_2^2), \partial_{\xi_2} g_1, \partial_{\xi_2}(g_2 \partial_{\xi_2} x_2)\}$

The first form of  $\mathbf{u}$  is a fairly general form, where the  $D^i$  operator is any admissible derivation over  $X$  of order  $i$ , including, for example,  $k$ th order ordinary derivatives,  $dx_i^k/d\xi_j^k$ , as well as  $k$ th order partial derivatives,  $\partial^k x_i / \partial x_j^k$ , etc. The second example is a very specific form, meant to search common terms found in the popular classical equations, here we consider terms from the one dimensional Navier-Stokes PDEs, where  $g_1(\mathbf{x})$  and  $g_2(\mathbf{x})$  are constitutive equations that encode the pressure and viscosity as some nonlinear function of the state variables, e.g. [**empty citation**]. The possibilities for searching admissible forms of  $\mathbf{u}$  are extensive, and will be explored in significantly more detail below.

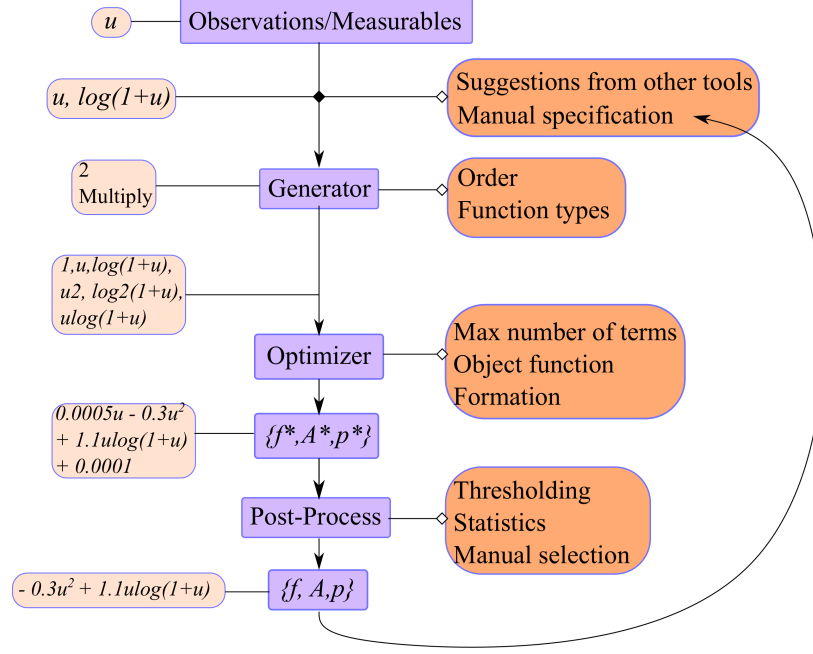


Figure 2: The general flow chart of the proposed SIR package.

### 2.1. Differential equation systems of the form $\frac{d\mathbf{x}}{d\xi_i} = \mathbf{a}\mathbf{f}(\mathbf{u})$

For simplicity of presentation, consider an equation of one independent variable  $\xi = \xi_1$ , e.g., for time domain solutions say  $\xi = t$ . Then given observables  $\mathbf{x}(\xi) = \{x_1(t), x_2(t), \dots, x_m(t)\}$ , we are trying to fit a differential system of the form:

$$\begin{bmatrix} \frac{dx_1}{dt} \\ \vdots \\ \frac{dx_m}{dt} \end{bmatrix} = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} f_1(\mathbf{u}) \\ \vdots \\ f_n(\mathbf{u}) \end{bmatrix}. \quad (1)$$

Here,  $\mathbb{X}_t = [\frac{d\mathbf{x}}{dt}]$  is a  $m$  by  $t$  matrix, and the right hand side of Eq. (1) uses the standard matrix product. The matrix  $\mathbb{A} = [a_{ij}]$  is an  $m \times n$  matrix of constants to be solved for, and the  $n$   $f_i(X)$  denote the  $n$  (potentially highly nonlinear) encodings, for example,  $f_i(\mathbf{u}) = x_1^2 \sin(x_2)$ . Together, these stack into the  $n \times t$  matrix  $\mathbb{F}_{\mathbf{u}} = \mathbf{f}(\mathbf{u})$ , such that symbolically the equation can be rewritten as

$$\mathbb{X}_t = \mathbb{A}\mathbb{F}_{\mathbf{u}}. \quad (2)$$

The least square solution of Eq. (2) can be obtained using the Moore-Penrose pseudoinverse of  $\mathbb{F}_{\mathbf{u}}$ ,

$$\mathbb{A} = \mathbb{X}_t \mathbb{F}_{\mathbf{u}}^+. \quad (3)$$

### 2.2. Generating possible terms

Significant freedom resides in constructing possible forms for  $\mathbf{f}(\mathbf{u})$ . At this level, three kinds of input are required from user to make the problem feasible:

1. Suspected measured quantities to be involved:  $x_1, x_2, \dots, x_m$ .
2. Suspected forms that may be involved:  $f_1(\mathbf{u}), f_2(\mathbf{u}), \dots, f_n(\mathbf{u})$ .

3. Some nonnegative integer  $n_\ell \in \mathbb{N}$  specifying the maximum degree allowed for the terms in the combinatorics.

As an example, consider the case  $\mathbf{x} = x_1, x_2$  with suspected function list  $x \mapsto x, x \mapsto \sin(x)$  and combination order up to  $n_\ell = 2$ . Under these conditions, the pool of terms to draw from include:

- Order 0 terms:  $C$
- Order 1 terms:  $x_1, x_2, \sin(x_1), \sin(x_2)$
- Order 2 terms:  $x_1 x_2, x_1 \sin(x_1), x_1 \sin(x_2), x_2 \sin(x_1), x_2 \sin(x_2), \sin(x_1) \sin(x_2)$

where  $C \in \mathbb{R}$  is some constant.

### 2.3. Rational Regression $\mathbf{y} = \frac{\mathbb{P}(\mathbf{u})}{\mathbb{Q}(\mathbf{u})}$

#### 2.3.1. Exact fit

As is the case in many application areas, it is interesting to know how an observed target signal  $\mathbf{y}$ , or some quantity of interest, behaves as a function of other observational signals  $\mathbf{x}$ . For this case, along with the case where  $\mathbf{y} = \mathbf{0}$ , and exact rational regression can be performed by solving,  $\mathbf{y} = \frac{\mathbb{P}(\mathbf{u})}{\mathbb{Q}(\mathbf{u})}$ . Numerically this corresponds to:

$$\mathbf{y}_{1 \times t} \odot (\mathbf{N}_{1 \times n} \mathbb{Q}_{n \times t}) = (\mathbf{M}_{1 \times m} \mathbb{P}_{m \times t}) \quad (4)$$

for  $\odot$  element-wise multiplication, and  $\mathbf{M}, \mathbf{N}$  the coefficient vectors to be determined. The forms of the  $\mathbb{P}, \mathbb{Q}$  are the admissible nonlinear basis terms from  $\mathbf{u}$ , and  $(\mathbf{N}_{1 \times n} \mathbb{Q}_{n \times t})$  and  $(\mathbf{M}_{1 \times m} \mathbb{P}_{m \times t})$  are the resulting linear combinations of these terms. Note that Eq. (4) can also be rewritten as the block matrix:

$$\begin{bmatrix} \mathbf{M} & -\mathbf{N} \end{bmatrix} \begin{bmatrix} \mathbb{P} \\ \tilde{\mathbb{Q}} \end{bmatrix} = \mathbf{0},$$

where  $\tilde{\mathbb{Q}}_{n \times t} = \mathbb{Q}_{n \times t} \text{Diag}(\mathbf{y})_{t \times t}$ , and the vector  $\mathbf{L} = \begin{bmatrix} \mathbf{M} & -\mathbf{N} \end{bmatrix}$  lives in the left null space of the matrix  $\begin{bmatrix} \mathbb{P} \\ \tilde{\mathbb{Q}} \end{bmatrix}$ .

#### 2.3.2. Approximated fit

In some applications instead of an exact fit—where the existence of solutions is either rare or the expression is too complicated to be physically interpretable; or, for that matter, intuitively meaningful—an approximated fit is often more pragmatic. In this context an approximate solution can be readily obtained by solving the following minimization problem:

$$\min_{a,b} \mathcal{M} \left( \mathbf{y}, \frac{a_1 \mathbf{P}_1(\mathbf{u}) + \dots + a_m \mathbf{P}_m(\mathbf{u})}{b_1 \mathbf{Q}_1(\mathbf{u}) + \dots + b_n \mathbf{Q}_n(\mathbf{u})} \right), \quad (5)$$

with constraints:

$$a_1^2 + \dots + a_m^2 + b_1^2 + \dots + b_n^2 = 1, \quad \text{and} \quad b_1^2 + \dots + b_n^2 \neq 0.$$

Note that  $\mathbf{P}_i$  and  $\mathbf{Q}_j$  are vectors of length  $t$ , corresponding to the columns of  $\mathbb{P}$  and  $\mathbb{Q}$  from section 2.3.1, and  $\mathcal{M}$  is an arbitrarily defined user-specified metric for measuring the quality of the fit. In this formulation, the user needs to specify required inputs, as described in section 2.2, for both the numerator and denominator.

#### 2.4. Dealing with combinatorial explosion

As is readily apparent from section 2.2, given  $m$  observables,  $n$  number of forms, and the maximum degree on the combinatorics  $n_\ell$ , the available number of combinations available to the algorithm becomes,

$$C(n, m, n_\ell) = \sum_{i=0}^{n_\ell} \binom{1 + nm}{i}. \quad (6)$$

The resulting number of combinations available provides significant freedom for finding a parsimonious fit to the data, but it also brings numerical difficulties, in that the number of admissible formulations grows rapidly. In consideration of trying to achieve a balance between the combinatorial explosion of possible expressions involved, and the fact that simpler forms are more convenient, and generally dogmatically preferable (e.g. Occam's razor, etc.) for human interpretation, we present several options available to reduce the resulting search.

##### 2.4.1. Constraining the formation

Instead of solving Eq. (5) that results in a combinatorial search of degree  $nm$ , we can force various hard constraints over the search domain to reduce the computational complexity. For example, the denominator degree can be fixed for each search round, such that with the denominator fixed as a constant, Eq. (5) becomes

$$\min_a \mathcal{M}(\mathbf{y}, a_1 \mathbf{P}_1(\mathbf{u}) + \dots + a_m \mathbf{P}_m(\mathbf{u})). \quad (7)$$

Similarly, if the user's domain knowledge provides that a specific form must appear in the denominator  $d(\mathbf{u})$ , then the relevant minimization problem becomes

$$\min_a \mathcal{M}(\mathbf{y}d(\mathbf{u}), a_1 \mathbf{P}_1(\mathbf{u}) + \dots + a_m \mathbf{P}_m(\mathbf{u})). \quad (8)$$

In this way, prior knowledge of admissible solution forms, or simplifying assumptions that can be utilized, will substantially decrease the search space as well as the rate in which the nonlinear order of the solution grows in combinatoric complexity.

Various other tricks can also be employed to similar effect. For example, if one adds the mapping  $\eta_i \mapsto \frac{1}{\eta_i}$  as an independent or dependent basis variable in  $\mathbf{u}$ , then ideal solutions that appear as  $\frac{\eta_i + \eta_j}{\eta_i \eta_j}$  in Eq. (5) can also be discovered as  $(\eta_i^{-1} + \eta_j^{-1})$  from Eq. (7), etc.

##### 2.4.2. Greedy Search

Even given constraints constructed along the lines in section 2.4.1, the computational complexity for an exhaustive search can grow rapidly, and become prohibitive. For example, with the formation in Eq. (7), there are still of order  $\mathcal{O}\left(\binom{C}{m}\right)$  possible expressions to compute and compare in this case as noted in Eq. (6).

In order to reduce the computational effort, one can employ a simple greedy algorithm. In this particular setting, the procedure would be as follows:

1. On the first iteration, select only one term from  $C(n, m, n_\ell)$  as the  $\mathbf{P}_1$  that gives the least error (i.e. smallest value of  $\mathcal{M}$ ) among all.
2. At the  $i$ th iteration select a  $\mathbf{P}_i$  from the remaining  $(C(n, m, n_\ell) - i)$  terms so that  $\sum_{j=1}^i a_j \mathbf{P}_j(X)$  is minimized.
3. Recursively perform the previous step until reaching the maximum  $m$  specified by the user.

Convergence of this greedy algorithm is readily achieved since the expression at iteration  $i$  is included in the expression at iteration  $i + 1$ , so that the absolute error is always decreasing with an apparently absolute lower error bound at zero. The greedy algorithm as mentioned above will require about  $\mathcal{O}(mC)$  regression calculations. Of course, the output of the greedy algorithm may very well not be the absolute minimum and the expression that results in the absolute minimum may not be unique. In practice, one can add randomness for the first two steps and compare the final output for selection and for proceeding to next steps. The reduced computational complexity also make things like deciding the appropriate value of  $n_\ell$  and  $m$  possible by using the “elbow method” from error vs.  $(n_\ell, m)$  plots. The ideal expression should not be too complicated to interpret.

## 2.5. Fitting with $\mathbf{f}(\mathbf{u}) = \mathbf{0}$

In many applications, one is not given a target signal  $\mathbf{y}$ , but instead is given measurable signals  $\mathbf{x}(t) = \{x_1(t), \dots, x_m(t)\}$  with the goal of discovering an implicit formula  $\mathbf{f}$  such that  $\mathbf{f}(\mathbf{u}) = \mathbf{0}$ . For cases such as these, we can simply add the constraints

$$a_1 + \dots + a_m = 1, \quad (9)$$

to Eq. (7) to avoid the trivial solution, while setting  $\mathbf{y} = \mathbf{0}$ .

### 2.5.1. Fitting one formulation with multiple signals

In many applications, observations are significantly polluted with either experimental uncertainty or an absence of observational determinism, making deterministic solutions to such real world systems of limited utility. In this context, the goal is to discover stochastic systems of the forms previously discussed, that are able to predict complex system behaviors that incorporate and predict the observed uncertainties present, and do so in as parsimonious a way as possible.

One way of approaching this problem is by noting that the formulation corresponding to Eq. (9) makes it very easy to adopt approaches such as Bayesian ridge regression (BRR) [[empty citation](#)] to perform statistical inference on the coefficients  $a_i$ , in order to recover stochastic forms of the resulting solutions. In this case, instead of specifying  $\mathbf{y} = \mathbf{0}$ , it is necessary to specify a prior distribution on the uncertainty of the residual error; for example,  $\mathbf{y}_i \sim U(-1, 1)$  as a uniformly distributed random variable between -1 and 1, etc. However, because of the significant sensitivities these methods have on the choice of prior used [[empty citation](#)], and due to the difficulty/expense of imposing and analyzing fully bayesian posteriors in order to determine the most appropriate prior to use on the residual [[empty citation](#)], these approaches can be either prohibitively expensive or, alternatively, rather brittle relative to user-specified priors [[empty citation](#)].

An alternative to the BRR approach can be utilized when the application scenario is enriched with multiple, e.g.  $J$ , samples of a given experimental instance  $\{\mathbf{x}_1, \dots, \mathbf{x}_J\}$  tracking the same physical phenomenon. Here the goal becomes to generate a single explicit stochastic form  $\mathbf{f}$  that is able of summarizing the observed behavior of the input sample distribution, where clearly the larger  $J$  becomes, the more effective the model becomes.

Here, using Eq. (7) for example, coefficients  $\{a_i\}$  can be extracted as random variables instead of constants. That is, set:

$$f_j = \min_{a_j} \mathcal{M}(y_j d_j(\mathbf{u}), \quad a_{1j} \mathbf{P}_{1j}(\mathbf{u}) + \dots + a_{mj} \mathbf{P}_{mj}(\mathbf{u})). \quad (10)$$

An  $f_j$  is solved for over every observed sample, and then, the following optimization is performed:

$$\min_{\mathbf{f}} \mathcal{N}(f_1, \dots, f_J), \quad (11)$$

where  $\mathcal{N}$  is now a measure for encoding the summarized residual error across all gathered samples. For example, if  $\mathcal{N}$  is the sum of the absolute residual with equal weights, then the optimization problem in

Eq. (11) would become:

$$\min_{\mathbf{f}} \sum_{j=1}^J |f_j| \quad (12)$$

Once the expression is fixed, the coefficients  $a^j = \{a_{1j}, \dots, a_{mj}\}$  can be calculated for each sample and used for making direct statistical inference. That is, for example, they may be binned and then a KDE [**<empty citation>**] may be constructed from the resulting histogram.

## 2.6. Validation examples

Generating validation examples is now easy using the SIR Package, and we have tested many such cases. In the deterministic setting the solutions are immediate, and follow many of the system identification [**<empty citation>**] and symbolic regression [**<empty citation>**] type behaviors that have been previously explored, with the added capability of encoding partial and arbitrarily high order derivatives, in the context of rational forms.

The SIR Package ...

However, one of the more intriguing aspects of the method has to do with being able to encode stochastic differential systems, which is where we focus the final # simple validation demonstrations below.

### 2.6.1. Lorenz system

As a demonstration of a classical chaotic subsystem, we consider the three dimensional Lorenz system taking the form:

$$\frac{dx}{dt} = \sigma(y - x), \quad (13a)$$

$$\frac{dy}{dt} = x(\rho - z) - y, \quad (13b)$$

$$\frac{dz}{dt} = xy - \beta z, \quad (13c)$$

with initial condition  $(1.0, 1.0, 1.0)$  and parameters  $\rho = 28.0$ ,  $\sigma = 10.0$  and  $\beta = \frac{8}{3}$ . The solution is sampled, as seen in Fig. 3, on the temporal grid  $t \in [0, 40]$  using  $\Delta t = 0.01$ .

The goal of this validation example is to recover system of ODEs in Eqs. (13) using the  $\frac{d\mathbf{x}}{d\xi_i} = \mathbf{a}\mathbf{f}(\mathbf{u})$  formulation, and given input  $\mathbf{x} = (x, y, z)$ . As a direct consideration, we can try first estimating derivatives of the input using 3-point central differencing schemes, select  $*, +, -$  for possible relations among terms and maximum order up to 3 *Craig says: how does this work out?* .

The output of the SIR package, given these computationally simple settings, surprisingly yields:

$$\frac{dx}{dt} = -9.98x + 9.98y, \quad (14a)$$

$$\frac{dy}{dt} = 27.81x - 0.96y - 0.99xz, \quad (14b)$$

$$\frac{dz}{dt} = -2.66z + 0.99xy. \quad (14c)$$

The package predicts the equation structure surprisingly well, with the estimated coefficients having an error of  $\mathcal{O}(\Delta t)$ . The thing to note here, is that a critical component in controlling the error seen in Fig. 4, it turns out, is in the estimation of the derivative (in this example, the 3-point central differencing scheme). This is generally the case, and seems to grow more sensitive as a function of derivative order, as expected. What is notable is how well this relatively crude and extremely computationally efficient finite difference approximation is able to match the solution.

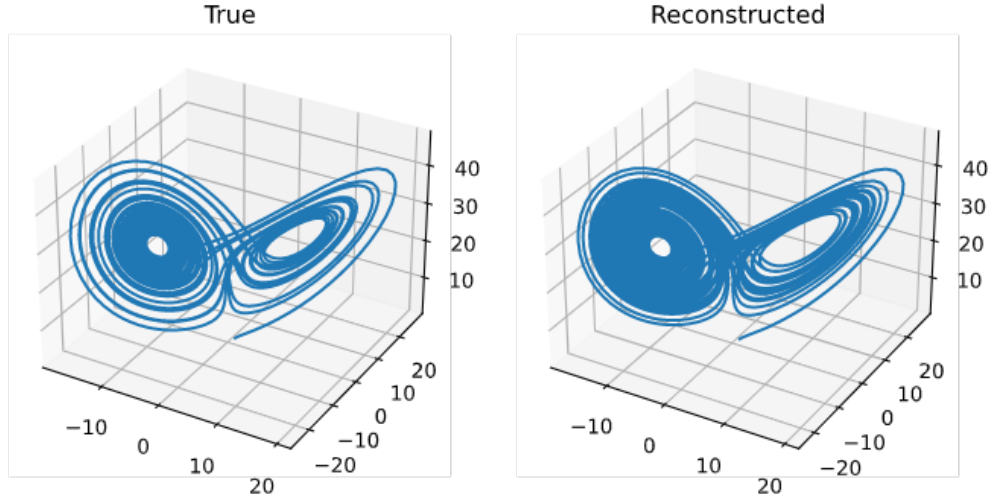


Figure 3: The ideal Lorenz system from Eq. (13) and the solution solved from Eq. (14) as predicted by SIR Package selecting a simple central differencing scheme as an estimation for the derivatives, with no preprocessing on  $\mathbf{x}$ .

The solutions of Eqs. (14) are demonstrated in both Figures 3 and 4, where they are compared with the original sampled signal, in some regards. That is, note that rather important qualitative features of the system are reasonably well-preserved. For example, being a classically chaotic system, the attractors of the dynamical system are preserved to convincing affect in Fig. 3. However, as again demonstrated in Fig. 4, the subsolution dynamics are fairly wonky. This is also, in part, because solutions to the Lorenz system are sensitive in their coefficients, and bifurcate along them in some sense.

In any case, the SIR Package, can readily utilize any approximations provided to it for computing the “jet bundles” of any complex, highly oscillatory, and/or pointwise discontinuous inputs. In this example the input Lorenz data is not preprocessed at all, and then a naive 3-point finite differencing method is used on it. However, it is relatively obvious how approximations of the derivatives on the raw data can be made as accurate as the user wants, e.g. by using higher order finite difference approximations. In addition, various other function space approximations can be recovered by setting stringent thresholds on regression preprocessing steps; for example, to achieve  $L^1$ ,  $W^{k,p}$ , etc. function approximations. Note that currently we can also always find smooth (i.e.  $C^\infty$ ) approximations to any set of signals to arbitrary, i.e. user-defined, accuracy. However finding precise, computationally efficient function space approximations for arbitrary input data is not entirely trivial, so we will save the details underlying this particular set of tools for another paper.

In any case, for the Lorenz system, using the exact derivatives defined in Eqs. (13)—in the sense of a classically manufactured solutions [**<empty citation>**]<sup>1</sup>—instead of the rough central differencing scheme used above, and leaving the other settings unchanged, recovers the system perfectly:

$$\frac{dx}{dt} = -10.00x + 10.00y, \quad (15a)$$

$$\frac{dy}{dt} = 28.00x - 1.00y - 1.00xz, \quad (15b)$$

$$\frac{dz}{dt} = -2.67z + 1.00xy. \quad (15c)$$

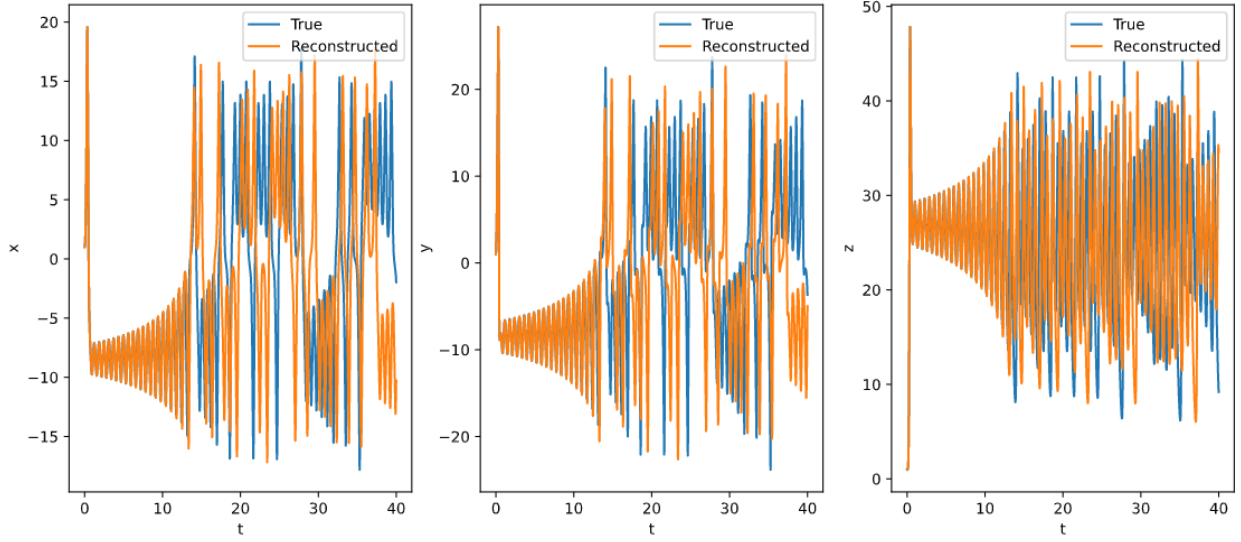


Figure 4: Comparison between the ideal Lorenz system components Eq. (13) and the solution of system Eq. (14) along each direction, with no signal preprocessing, and using the crudest first order finite difference approximation. The result is first order accurate,  $\mathcal{O}(\Delta t)$ , however with nontrivial artifacts in the components.

### 2.6.2. Harmonic Oscillator with Random Coefficient, e.g. $\mathbf{y} = \mathbf{0}$

Consider the observed data generated by  $h(\mathbf{x}) = \sin(\alpha\pi\mathbf{x})$  where  $\alpha \sim U[1, 2]$  is a random variable drawn from a uniform distribution between  $[1, 2]$ . The grid generating  $\mathbf{x}$  has 100 points, and we draw 40 samples of  $\alpha$ ,  $m = 40$ , as shown in Fig. 5. These  $m$  samples are then used for training, where in addition, the first and second derivatives are utilized by taking simple finite difference approximations on the function  $h(\mathbf{x})$ , though we will discuss utilizing other—more robust—options below. The resulting triple  $\{h, dh/dx, d^2h/dx^2\}$  is then trivially computed, and a search is performed over  $n_\ell = 2, n = 3$ .

The function  $h$  is of course a solution to the simple harmonic oscillator,  $d^2\theta/dt^2 + C\theta = 0$ , when  $x = t$ , and thus we would expect this solution to be found in our search, with relatively low error. Indeed, we find just that. The first solution in the parsimonious greedy search that shows low error takes the form over its maximum likelihood of:  $c^{-1}(ah_{xx} + bh) = 0$ , for  $a = 0.018198, b = .397015, c = -.397492$ , which, again, is a form of the harmonic oscillator equation when  $\mathbf{x} = \mathbf{t}$ . For the resulting distributions on the coefficients,  $a, b$ , and  $c$  can be collapsed into a single random variable and rescaled by  $\pi^2$  as shown in Fig. 5, showing that the distribution found by SIR matches almost exactly the one drawn from the original samples  $\alpha$ , and propagates through to find an entirely admissible, and readily parsimonious system representing the function  $h(\mathbf{x})$ .

Of course, there are many equations that are slightly, or significantly, more complex that also match the observed data, as shown in Table 1. An interesting insight that SIR provides (as also does, for example, the Method of Manufactured solutions [[empty citation](#)]) is how enormously degenerate differential systems are, representationally, for even simple observational functions such as  $h$ . This leads to a number of interesting considerations, some of which will be discussed in more detail later.

### 2.6.3. Stochastic ODE with Noise e.g. $\mathbf{y} = \mathbf{f}(\mathbf{x})$

The same basic approach as in section 2.6 can be applied to finding generating stochastic equations,  $\mathbf{y} = \mathbf{f}(\mathbf{x})$ , instead of searching for an equation that matches a generating stochastic function, as in section 2.6. For this example the solution method is shown using Bayesian ridge regression (BRR) [[empty citation](#)], where all relevant priors are arbitrarily set to 1e-6 by default.

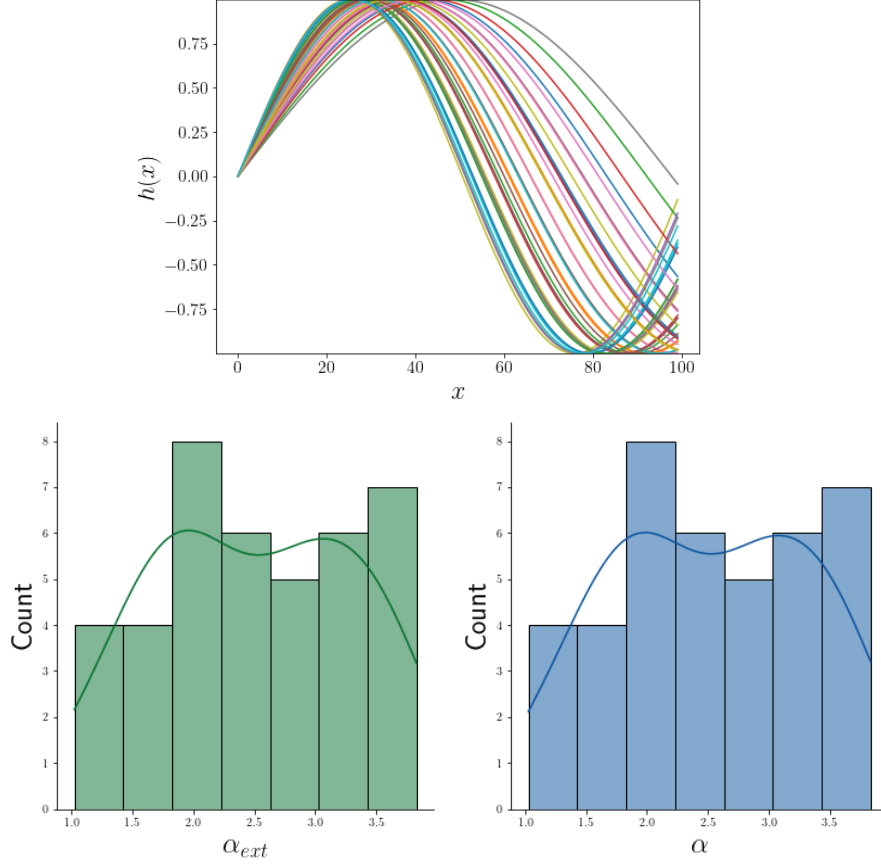


Figure 5: Forty samples from the generating function  $h(\mathbf{x})$ . The distribution on the bottom left is recovered using the  $m$  ( $b/a\pi^2$ )’s while that on the bottom right is constructed using the uniformly drawn  $m$  samples  $\alpha^2$ .

For a generating function we consider

$$\mathbf{y} = h(\mathbf{x}) + \alpha h'(\mathbf{x}) + \zeta \quad (16)$$

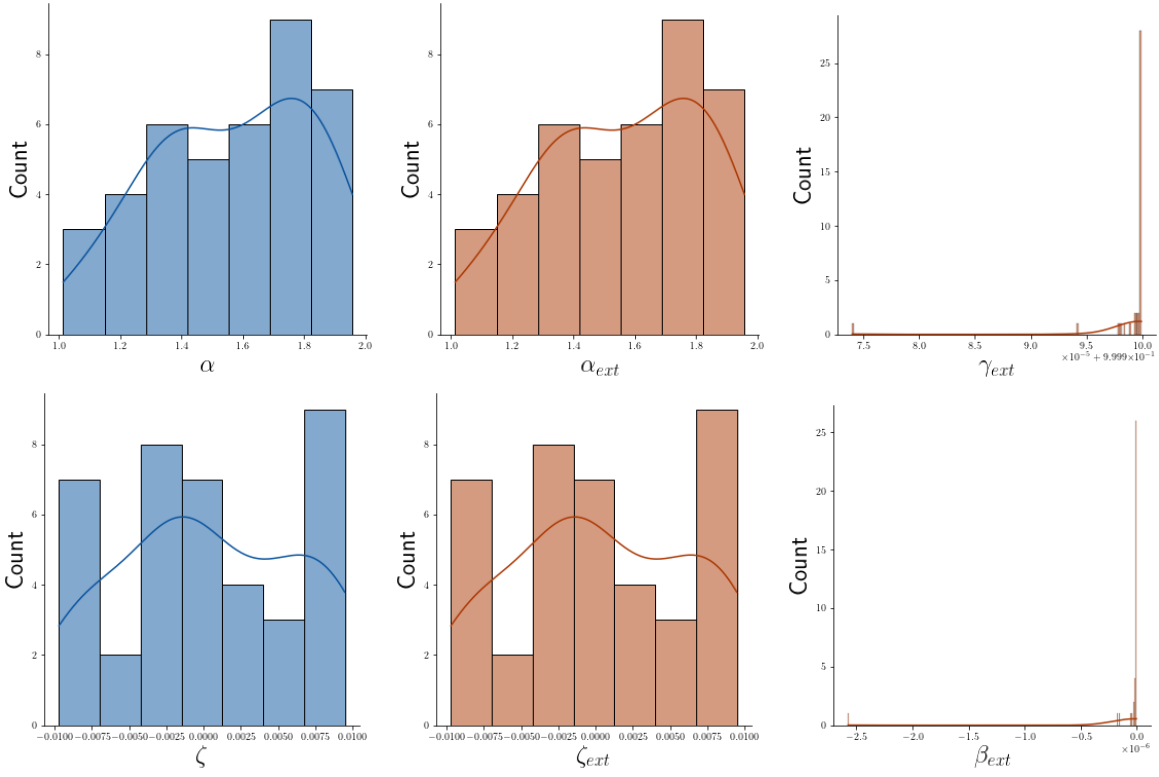
where  $\alpha \sim U[1, 2]$ , and  $\zeta \sim U[-1e-2, 1e2]$ . The same grids as in section 2.6 are used here again, with  $m = 40$ ,  $n_\ell = 2$ ,  $n = 3$ . Again the greedy algorithm is used, and again the resulting solutions rapidly converge to a representational facsimile of the generating equation, in this case finding

$$\mathbf{y} = \gamma_{ext} h(\mathbf{x}) + \alpha_{ext} h'(\mathbf{x}) + \zeta_{ext} - \beta_{ext} h''(\mathbf{x}). \quad (17)$$

Note here, as shown in Fig. 6, that the distributions on the extracted coefficients  $\alpha_{ext}$  and  $\zeta_{ext}$  match the generating distributions  $\alpha$  and  $\zeta$  almost exactly, and the extracted distribution  $\gamma_{ext}$  is strongly peaked, i.e. mean value of 0.999999 to single precision, near the ‘true’ value of one. The algorithm does infer an extremely small, i.e. mean value of zero to single precision, highly peaked coefficient  $\beta_{ext}$  on the second derivative as well. Note that this solution has an  $L^2$  matching error between Eq. (16) and Eq. (17) of 3.9328e-09.

Table 1: Admissible Forms returned from Greedy Algorithm

| $L^2$ Error | Numerator Form                          | Denominator | Std in Coeffs                 |
|-------------|-----------------------------------------|-------------|-------------------------------|
| 345.341     | $0.707d_{xx}h$                          | -0.707      | [4.4e-16,4.4e-16]             |
| 0.0016      | $0.018 d_{xx}h + 0.397 h$               | -0.397      | [0.007, 0.007, 0.007]         |
| 0.0016      | $0.016 d_{xx}h + 0.369 h + 0.0001 d_xh$ | -0.369      | [0.006, 0.028, 0.0004, 0.028] |
| 0.3740      | $0.707 h^2$                             | -0.707      | [4.4-16, 4.4e-16]             |
| 0.0012      | $0.396 h^2 + 0.018 hd_{xx}h$            | -0.397      | [0.0059, 0.007, 0.006]        |
| 0.0011      | $0.346 h^2 + 0.015 hd_{xx}h + 0.004 h$  | -0.362      | [0.106, 0.009, 0.005, 0.022]  |


 Figure 6: The blue distributions on the left correspond to the  $m$  samples drawn for the generating  $\mathbf{x} + \alpha\mathbf{x}' + \zeta$ . while on the right are the distributions extracted by the SIR algorithm.

#### 2.6.4. Heat Equation with Stochastic Coefficient, $\mathbf{y} = \mathbf{0}$

The basic approach works just as well for discovering systems characterized by partial differential equations (PDEs), and for readily identifying degenerate subsolutions to those systems. In fact, the inclusion of partial derivatives, as we shall see later, is able to reveal interesting correlative dynamics in complex system behaviors. Here however, we show an additional simplistic validation example, where a stochastic solution to the heat equation is naively used as a generating function,  $g(\mathbf{x}, t) = \exp(-\alpha t) \sin(\mathbf{x})$ ; again setting  $\alpha \sim U[1, 2]$  and  $m = 40$ , in this case over a uniformly spaced 100 point grid  $\mathbf{x} = [0, \pi]$ , and a uniformly spaced temporal grid of 50 points over  $t = [0, 1]$

In this context we search a larger space, computing  $\{g, g_x, g_{xx}, g_t, g_{tt}\}$  again using  $n_\ell = 2, n = 3$ . While

one might expect the result to converge to a stochastic form of the heat equation, what we find instead is that this solution to the heat equation is actually deeply degenerate, and is more parsimoniously encoded (relative to our definition of parsimony) in many simpler differential equations. For example, the simplest immediately found by the greedy algorithm is:

$$g + \gamma_{ext}g_x = 0,$$

where the resulting parameter  $\gamma_{ext}$  simply numerically tries to approximate  $\tan(\mathbf{x})$ . It turns out that the SIR algorithm is able to easily find this approximate  $\gamma_{ext}$  to very high accuracy, yielding an  $L^2$  error of 6.059e-12.

Another degenerate solution form found is

$$g + \beta_{ext}g_{xx} = 0,$$

which similarly gives an  $L^2$  error of 8.7315e-08. This is obviously an approximate solution to the system, as the algorithm, and indeed the mathematics, cannot distinguish between the terms  $g$  and  $-g_t$ , or  $g$  and  $-g_{xx}$ . In these cases there is no priori reason that the SIR algorithm will choose one form over another, other than which form the greedy algorithm happens to find first. Indeed, there is no real reason, when solving a system with a generating function, such as  $g(\mathbf{x}, t) = \exp(-\alpha t) \sin(\mathbf{x})$ , to assume the heat equation, when actually the dynamics of the system are fully encoded by many simpler equations.

It turns out this is frequently the case, given ansatz functions for analytic solutions to particular PDEs, what we often find is that the subsolution is actually more parsimoniously described by often significantly simpler, more direct equations that are easier to understand and frequently just as, if not more, intuitive and physically insightful.

### 2.6.5. Misrepresentational equivalences – invariance of representations

In most of the previous validation examples, the package easily starts to converge to the unknown system used to generate  $\mathbf{x}$ , but as seen in the heat equation example in section 2.6.4, the potential systems found by the algorithm in a given context may not be unique.

As another simple example,  $u = e^{-t} \sin(x)$  satisfies more than one relation:

$$f_1 : \frac{\partial u}{\partial t} = -u, \tag{18a}$$

$$f_2 : \frac{\partial^2 u}{\partial x^2} = -u, \tag{18b}$$

$$f_3 : \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, \tag{18c}$$

$$\dots \tag{18d}$$

While  $f_1$  (capturing the time dynamic) or  $f_2$  (capturing the spatial distribution) alone may not be the relation one is looking for, a result including both  $f_1$  and  $f_2$  together implies  $f_3$  (capturing both time and spatial dynamics) by substituting  $u$ . In similar cases where multiple relations with different structures present in the output together with errors below tolerance and each one of them seems to only cover partial information, one can use them to construct further relations that have more desired properties. In the next section we will demonstrate how these types of techniques can be used to extract insight from observation.

### 2.7. Simple real world example: MEG signals in the motor cortex

We have applied SIR to many real world data sets, in a number of different application areas, but in keeping with the sensibility of this paper we consider the popular BCI competition IV, dataset 3 used for classifying directionally modulated MEG activity to create brain-computer interface (BCI) systems that sense the directionality of subject’s hand movements for motor intention recognition<sup>1</sup>. We further note the close relationship—which will become clearer throughout this section—MEG signal interpretation can have to that of transcranial magnetic stimulation (TMS), which is used, among many other things, for pain reduction [Att+21], neural rehabilitation [Min+08], motor function regeneration [Xie+21], and traumatic brain injury therapeutics [Pin+21; AP13]. For simplicity, we refer the reader to our previous paper [KM21] where we developed a state-of-the-art subject-independent end-to-end classifier system for this same data set, and for additional information on the data we refer the reader to the original study [Tan+12].

For convenience we provide a cursory overview of the dataset under consideration. The BCI competition IV, dataset 3 consists of MEG signals taken from two subjects ( $S_1$  and  $S_2$ ), both right-handed, performing wrist movements on a joystick in four different directions from a fixed center position. The four movements consist of continuous movements by the right hand, starting at the center resting position to one of four target positions: right (R), forward (F), left (L), and backward (B) [HS12].

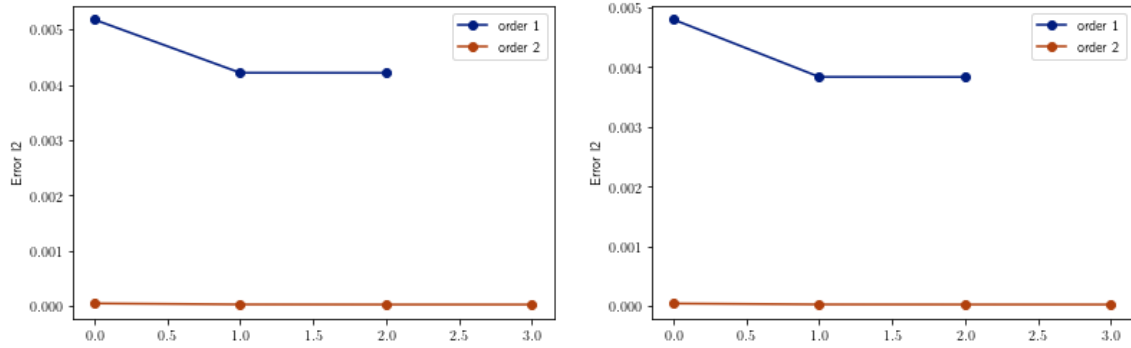
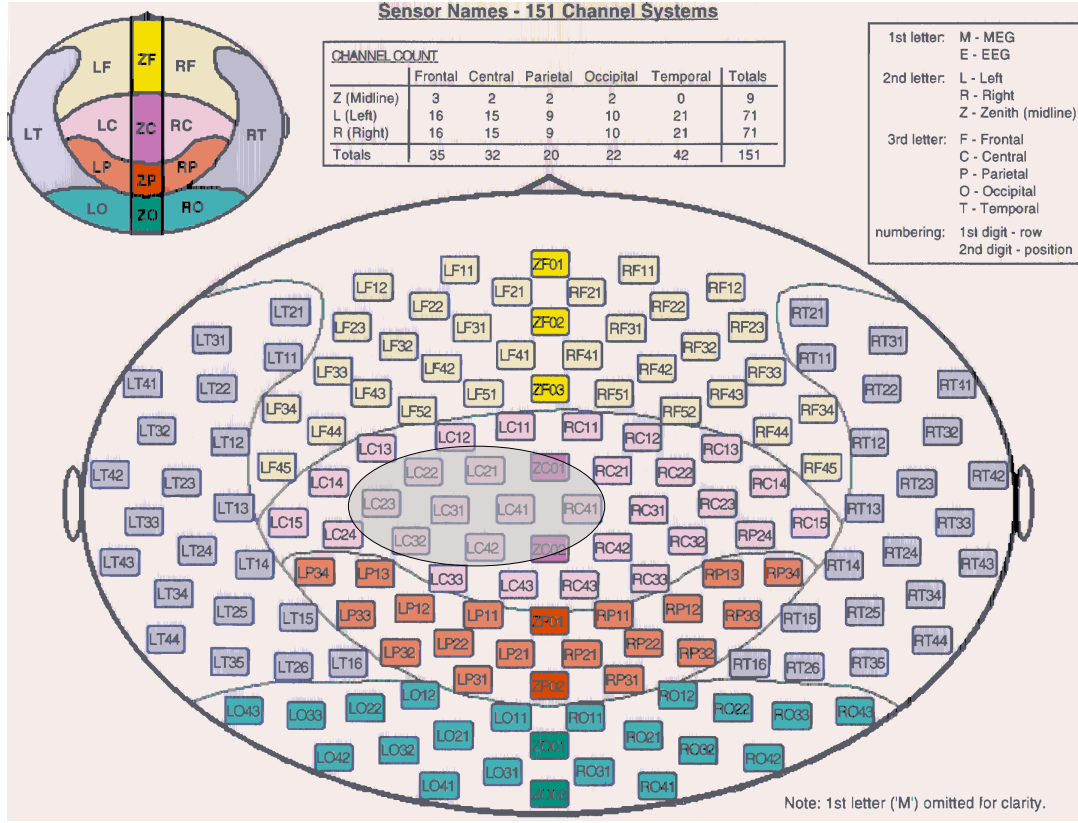


Figure 7: The  $x$  axis denotes the number of terms, where constant terms are not counted, and indexing starts at zero. The graphs show the  $L^2$ -error of Channel  $C = 3$  on the left for a single term second order fit is  $5.059\text{e-}05$ , and on the right for  $C = 8$  it is  $4.478\text{e-}05$ . In all channels the single term second order fit has error in the range of  $10^{-5}$ .

A total of ten MEG channels  $C = 0, \dots, 9$  are recorded from an array of 151 total sensors. The ten channels are selected to be relatively localized to the “motor cortex” anterior to the central sulcus, consisting of 7 sensors in the left central hemisphere, two sensors localized to the central zenith (e.g. Corpus Callosum), and one sensor located in the right central hemisphere, as shown in Fig. 8. Note that in this paper, the channels  $C$  are labelled, from channel 0 to channel 9, as: LC21, LC22, LC23, LC31, LC32, LC41, LC42, RC41, ZC01, ZC0. In previous work we developed classifiers able to extract the performed movement from the channel data, as well as to identify various other signature features of the MEG signals [KM21]. Here we provide a simple test and validation example for the SIR algorithm, to begin to show how it may be utilized in this type of context.

To start we can easily use SIR to examine the temporal relations implied and readily extracted from each channel individually. This leads to some initial observations. On a cursory SIR examination, it is

<sup>1</sup>We note that attempting to validate a cortex-wide system from biological neuron simulation models, such as the Hodgkin-Huxley model [empty citation], cable system [empty citation], or multicompartmental models [empty citation], while intrinsically of great interest, is both computationally prohibitive and wholly incomplete from a first principles perspective when trying to model the aggregate surface behaviors observed from an MEG experiment, for example.



VSM MedTech LTD, Vancouver

Figure 8: Locations of sensors used in the dataset: a 151-channel whole-head MEG system (VSM MedTech, Vancouver, British Columbia, Canada), reproduced with permission [Wal+08].

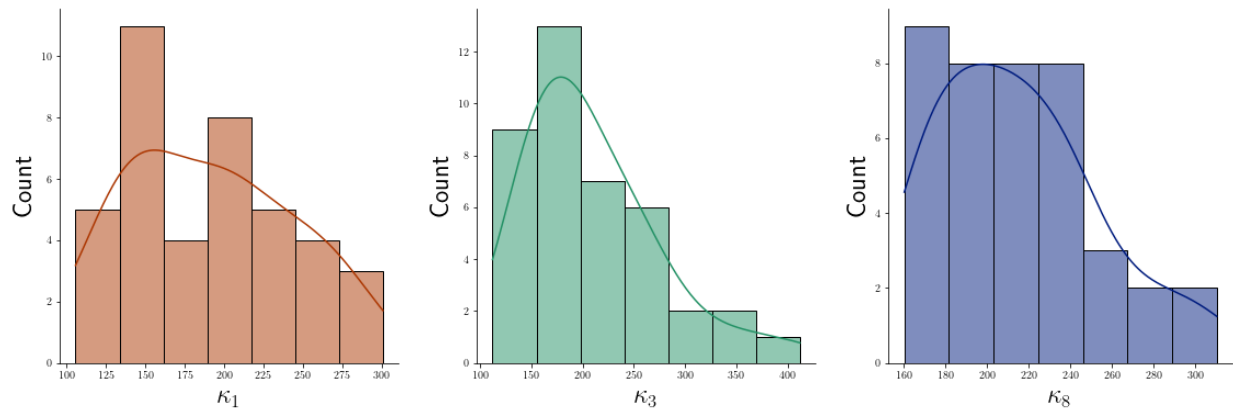


Figure 9: The  $\kappa$  distributions on three different channels for  $S_1$ , movement (R).

perhaps surprising that the magnetic fluctuations from the MEG signals imply remarkable consistency in the temporal frame, relative to each movement type. For example, utilizing the stochastic version of the  $\mathbf{f}(\mathbf{x}) = \mathbf{0}$  formulation from section 2.6.4 to examine each signal from subject  $S_1$ , we find that all movement types regress to parsimonious representations satisfying the same second order two term equation up to a one parameter random variable,  $\kappa$ :

$$\kappa_i \left( \frac{d^2 C_i}{dt^2} \right)^2 = 1, \quad (19)$$

where  $\kappa_i$  is a random variable drawn from a unique ratio distribution for each  $i$ th channel. Note that the square here on the second order term is an example of the types of degeneracy that can emerge, as discussed in sections 2.6.4 and 2.6.5. Also note that the distributions associated to the  $\kappa_i$  seem to encode the difference in the movement representations (R), (F), (L), and (B), the subjects  $S_j$ , as well as the uniqueness in each  $i$ th channel. To demonstrate this we examine two channels.

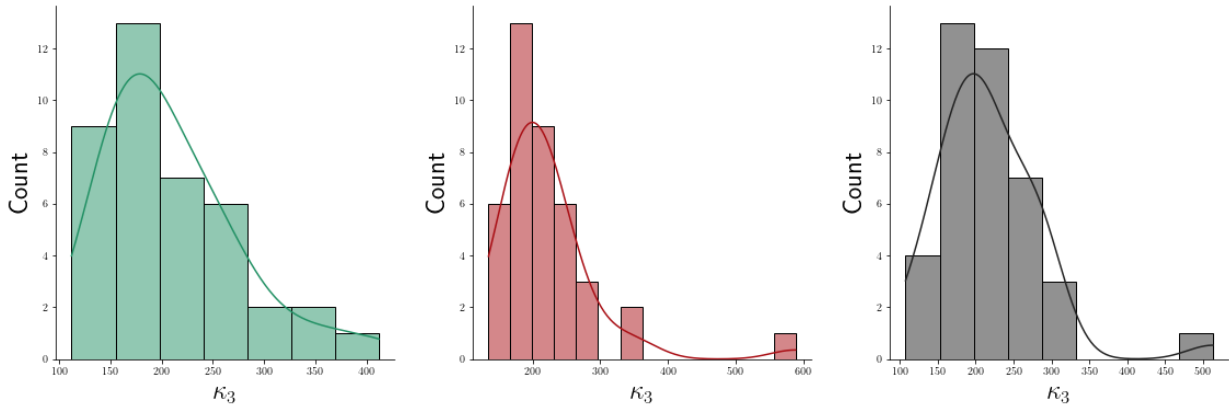


Figure 10: The  $\kappa$  distributions on channel  $C = 3$  for  $S_1$ , movements (R), (F), (L), in order from left to right.

In [KM21] channel  $i = 3$  is shown to carry significantly more “characteristic” signal relative to the movements of interest than the other channels, while channel  $i = 8$  is seen to be one of the “least important” relative to the classifier training regiment (e.g. see Fig. 6 from [KM21]). The data from the two channels nevertheless infer the same dynamical rates from the SIR fits, i.e.  $d^2 C_i / dt^2 = \kappa_i^{-1/2}$ , up to the distributions the random variables  $\kappa_i$  are drawn from.

To understand why this particular representation is chosen/discovered by SIR, we note from Figure 7 that the single term second order fit is chosen for all channels because it is significantly more accurate than all first order fits, and almost exactly as accurate as other second order fits with additional terms added. Consequently, utilizing a basic parsimony condition, the single term second order fit is chosen for all channels. In Fig. 9 we see the different distributions formed from both different channels for a single movement type, and in Fig. 10 over the same channel for different movements. The differences in these distributions are indicative of signature behaviors, suggesting that the ODE is sensitive to the coefficient scaling the temporal variation, on the one hand. On the other, it suggests a remarkably simple fact: that the signals “acceleration” with respect to each  $i$ th channel, i.e.  $d^2 C_i / dt^2 = \kappa_i^{-1/2}$ , is equal to a unique inverse distribution to surprisingly good accuracy (a temporal component reminiscent of many interesting dynamic equations, such as the acoustic wave equations and/or electromagnetic wave equations [Mic+16; MO96; MO97; OM97; Han08; Zas02]), e.g. see Fig. 11 for a comparison between the training and testing data distributions for  $\kappa_3$  over the movement types.

This mathematical observation has a simple and straightforward MEG-based interpretation:

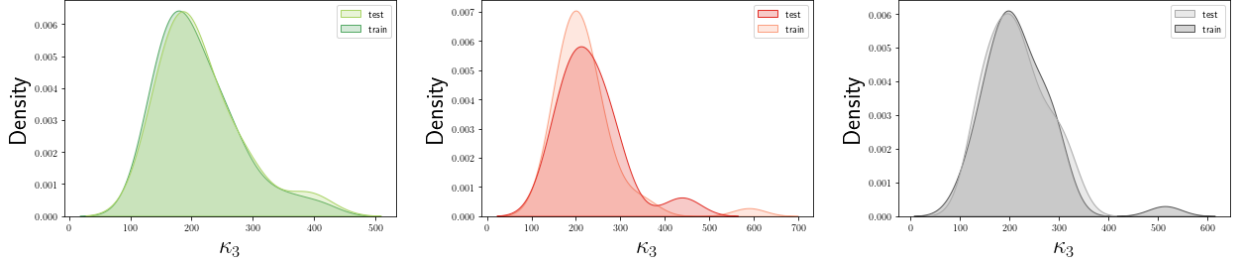


Figure 11: Comparison of the testing and training data distributions on  $\kappa_3$  for movement (R), (F), (L) in subject  $S_1$ , from left to right.

**Interpretation:** *the rate equations capture subtle temporal oscillations in the signal channels responsible for encoding the relative sensitivities stored parsimoniously in the corresponding coefficient distributions.*

It should be noted that BCI competition IV, dataset 3 is quite sparse—the testing set is comprised of only  $\sim 15$  samples per channel, and the training set only 40 samples per channel—yet even so, the testing and training distributions align fairly well up to potential outliers near and above  $\kappa_3 \gtrsim 400$ . The concerning statistical correlation between the testing and training sets for  $\kappa_3$ 's on the (F) movement densities (as seen in Fig. 11), for example, may be an indication of why even state-of-the-art classifiers struggle to achieve high accuracy [KM21]. This is in all likelihood in part due to sampling error caused from using such a sparse sample [LNM16], but it could also be an indication that Eq. (19) is too simplistic of a representation, and that the temporal components of the signals do not carry enough nuance to disentangle the important features latent in the MEG signal into functionally interpretable content. Nevertheless, a cursory examination of the KL-divergences between the output test and training distributions shown in Fig. 11, indicate a greater similarity between test and training sets of the same movement, than between test and training sets of different movement types, etc.

Broadly, at this stage, all this really tells us is that sensitivities along the variation in time of the MEG signals are able to uniquely capture signature features of the channel, and that one easy way of understanding those variations is as a single parameter distribution,  $\kappa$ , corresponding to the scaling of the signal acceleration. It is interesting to note here that the type of signal modulation generally used in, for example, transcranial magnetic stimulation, is generally of a fixed or arbitrarily varying frequency [Tur+21]. In the simplest sense, such a representation parallels the result from Eq. (19), in the deterministic sense, where a simple analytic and deterministic form of this system  $C''_i = a \sin(\pi b t)$  implies that  $C_i = -\frac{a \sin(\pi b t)}{\pi^2 b^2} + c_1 t + c_2$ , which is commensurate with current frequency and amplitude modulation practice, given constants  $a, b, c_1$ , and  $c_2$ . It is also striking to note that if we take the solution from a single realization of the stochastic Eq. (19) found by SIR, the solution to each realization—assuming no time dependence of  $\kappa$ —takes the form  $C(t) = \frac{a_1 t^2}{2} + c_3 t + c_4$ , which implies that random variables, and their corresponding distributions, exist such that the two formulations can be made equivalent; though not necessarily very easily stumbled across.

However, we can easily expand on this simple model in order to develop more insight into the observation space, and along with insight, increasingly more powerful tools. For example, If we now allow for spatial oscillations between the ten channels, indicative of phenomena like phase synchronization [PPK05] for example, we obtain an entirely different result with an entirely new set of insights into the data and the phenomena it represents.

As an example of this, the SIR algorithm can be applied under the assumption that the most characteristic signal channel is a function of not only time, but now of each of the other signal channels,  $C_3 = C_3(t, C_0, C_1, C_2, C_4, \dots, C_9)$ , which incidentally correspond to specific sensor locations, see Fig. 8. This leads to a phase space structure reminiscent of, e.g. total temporal or spatial derivatives (since each

channel is associated to a fixed location on the scalp, i.e. with MEG channels there is the additional implicit parametric dependency,  $C_i = C_i(\dots, \mathbf{x}_i)$ , when corresponding between channels  $\partial C_3(\dots, \mathbf{x}_3)/\partial C_i(\dots, \mathbf{x}_i)$  for  $i \neq 3$ .<sup>2</sup>

When such a procedure is performed on the data from subject  $S_1$  for movement (L), for example, there is a preference in the SIR algorithm for highly interactive second and third order representations, indicating rich and complex interactions and oscillations present between channel three and many of the other signal channels. The equations that emerge from the search yield:

### Parsimonious Fits with Three Terms

$$\begin{aligned} \text{Third Order Fit : } & \alpha_0 \left( \frac{\partial C_3}{\partial C_2} \frac{\partial C_3}{\partial C_8} \frac{\partial C_3}{\partial C_9} \right) + \beta_0 \left( \frac{\partial C_3}{\partial C_2} \frac{\partial C_3}{\partial C_7} \frac{\partial C_3}{\partial C_8} \right) + \gamma_0 \left( \frac{\partial C_3}{\partial C_1} \frac{\partial C_3}{\partial C_2} \frac{\partial C_3}{\partial C_5} \right) = \delta_0, \quad \text{err} = 0.0022, \\ \text{Second Order Fit : } & \alpha_1 \left( \frac{\partial C_3}{\partial C_2} \frac{\partial C_3}{\partial C_9} \right) + \beta_1 \left( \frac{\partial C_3}{\partial C_1} \frac{\partial C_3}{\partial C_8} \right) + \gamma_1 \left( \frac{\partial C_3}{\partial C_8} \frac{\partial C_3}{\partial C_9} \right) = \delta_1, \quad \text{err} = 0.0219. \end{aligned} \tag{20}$$

The errors (i.e. err implies  $L^2$  error) are reported on the right in Eq. (20). While the third order representation is an order of magnitude more accurate, both representations capture the phenomena to better than first order accuracy, and hence are both of functional interest.

In contrast to the temporal relations from Eq. (19), the distributions on each term encode signature nonlinear characteristics relating to the sensitivities of those terms in relation to the others. In this case, the distributions on the coefficients are quite interesting, almost all of which demonstrating strong multimodal characteristics, e.g. see Fig. 12, where all of the coefficients are of order unity, suggesting that each term is of “comparable” importance up to the sensitivity within the individual partial derivatives themselves (which can be easily examined).

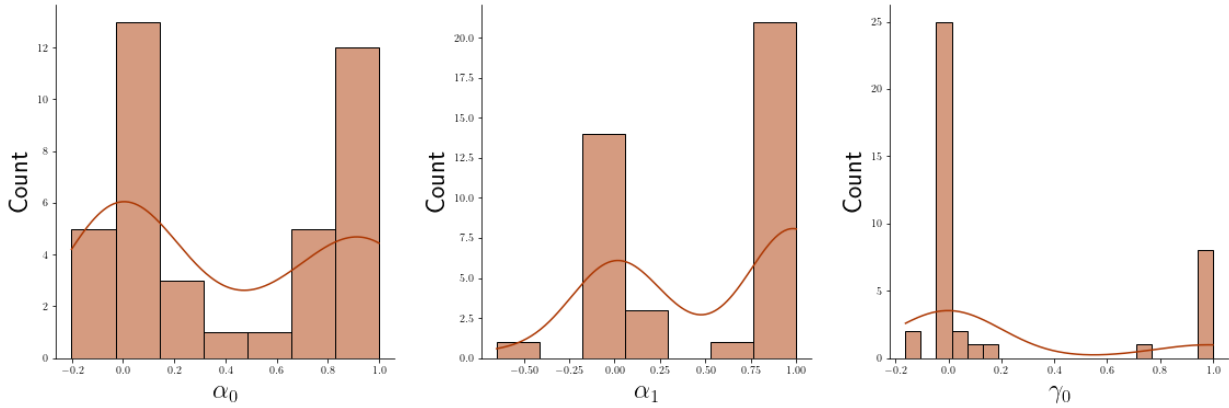


Figure 12: Example distributions from the spatial parsimony fits.

While the distributions seem to contain significant information, the individual terms lend themselves, in a fairly straightforward way, to direct interpretation. For example, the second order terms from Eq. (20), can be read in an MEG-based interpretation as being very similar to standard wave phenomena occurring over

<sup>2</sup>It should also be noted, as further discussed in section 2.6.1, that the choice of how one computes these derivatives can influence the resulting representation.

a curved manifold [RBL04]—where the curved manifold here represents the autopoietically correlated electrochemical and neurochemical pathways present in the cortex, that drive the electromagnetic observations embedded in the MEG signals. More plainly this simply means:

**Interpretation:** *In subject  $S_1$  highly resonant second order mixed oscillations occur between the variations of channel three with variations in channels two and nine, channels one and eight, and channels eight and nine, respectively.*

Similary for the third order terms:

**Interpretation:** *In subject  $S_1$  highly resonant third order mixed oscillations occur between the variations in channel three with variations in channels two eight and nine, channels two seven and eight, and channels one two and five, respectively.*

The spatial and temporal examinations can be performed to any differential order, to test for potential oscillation resonances present within channels, which in this case are resonances between locations in the motor cortex. Clearly this type of information can now be used to identify highly nonlinear resonances present during—and potentially involved in—hand movements. Particularly those hand movements fine-tuned to either particular subjects (in this case  $S_1$  and  $S_2$ ), or as we will see expounded on in section 3 in a slightly different context, to various demographic subgroups of collections of subjects, etc.

This type of procedure can then continue to be expanded by noting that the above also implies by construction that subject  $S_1$ 's MEG data for movement (L) is characterized by:

$$\frac{d^2 C_3}{dt^2} = \alpha_1 \left( \frac{\partial C_3}{\partial C_2} \frac{\partial C_3}{\partial C_9} \right) + \beta_1 \left( \frac{\partial C_3}{\partial C_1} \frac{\partial C_3}{\partial C_8} \right) + \gamma_1 \left( \frac{\partial C_3}{\partial C_8} \frac{\partial C_3}{\partial C_9} \right) - \delta_1 + \kappa_i^{-1/2}, \quad (21)$$

which indicates both a spatial and temporal variational correspondence that parsimoniously characterizes the time evolution of channel three, relative to the chosen context, and begins to develop a fairly insightful variational equation that is uniquely tuned to subject  $S_1$ . Of course, we can continue to simply add interesting dependencies, based on things we know about the data, or relational interactions we can easily infer from what we know about functional neuroimaging, neuroscience, electrophysiology, etc.

In any case, if we perform the same procedure on subject  $S_2$  for movement (L) we get a very similar accuracy of fit, though find a notably different set of inferred variational dependencies:

$$\frac{d^2 C_3}{dt^2} = \alpha_2 \left( \frac{\partial C_3}{\partial C_6} \frac{\partial C_3}{\partial C_8} \right) + \beta_2 \left( \frac{\partial C_3}{\partial C_1} \frac{\partial C_3}{\partial C_5} \right) + \gamma_2 \left( \frac{\partial C_3}{\partial C_1} \frac{\partial C_3}{\partial C_4} \right) - \delta_2 + \kappa_i^{-1/2}, \quad (22)$$

where here, not only do the coefficient distributions carry different characteristic information, but the strongest underlying spatial resonances themselves are altered. The basic interpretation of this is:

**Interpretation:** *Subject  $S_1$  and subject  $S_2$  have notably different second order oscillation resonances in the spatial variations characterizing the time evolution of channel  $C = 3$ .*

Clearly there is significant utility in having such representations for both identifying/classifying and analyzing/understanding motor intention in subjects and groups of subjects. The advantage of using a SIR model to using a straight classification model [KM21] for such a straightforward task, is that SIR extracts significant causal and correlative insights alongside its predictive capacity, that can reveal additional functional information, and do so to indefinite degrees of depth. Of course using both alongside each other also has advantages.

This type of functional dynamic information can also be easily utilized for application in neuronal entrainment studies [Her+16], for example, where not only is the parsimonious model so simple and compact that it can trivially run in real time, but it can also be updated and adapted to new incoming data in

real time. These types of simple formulations can be used to develop highly targeted modulation regiments for therapeutics using things like transcranial magnetic stimulation (TMS) and digital oscilloscopes, for example, allowing for complex, nonlinear, dynamical, and highly targeted approaches. These can be used in place of far more coarse and generic frequency/amplitude modulation approaches [Tur+21] that are difficult/impossible to fine-tune to individual subjects and/or groups, and are challenging to turn into highly targeted biofeedback mechanisms.

While the equations that emerge from these types of SIR procedures can be used as good candidates for efficient prediction, control, and analysis through brain and human computer interface technologies and neural oscillation therapeutics, perhaps their greatest utility lies rather in the ability to expand their scope indefinitely to include increasingly more integrated characteristics of complex systems, and to do so dynamically. However, the example in this subsection is just for simple low-level demonstration purposes. We will save a more fully realized application of these methods to the development of efficient biomedical devices and analysis tools in the motor cortex for another time.

### 3. System Architectures for Integrated Analysis (SAFIA)

#### 3.1. Abstract framework

The power of SIR systems of representation are not so much that they produce better, or more accurate representations of particular observed subsystems. While this seems to be true in some cases, in many circumstances there remain easy low-hanging arguments that the resulting encoding/equation is either not unique, not sufficiently parsimonious, or not sufficiently physically interpretable, and/or some mixture of the three. This is frequently the case, and indeed, one of the insights that the system exposes is just how incredibly degenerate systems of differential equations, for example, actually can be when matched to real-world observations. Rather, the power of SIR systems comes from being able to encode complex phenomena into incredibly efficient, rapidly computed, parsimonious (enough) representations, which then can be made easily human interpretable and then recursed over and made systematic and functorial.

These representations can then be used to generate system architectures for integrated analysis (SAFIA). The resulting SAFIA can be automated and dissected for complex system analysis, optimization, feedback, control, etc. They can be very efficient at encoding massively complex data taxonomies, and can be made unconditionally stable numerically; which here means they do not “blow up” or become uninterpretable due to cascading inconsistencies spread throughout symbolic tiers. They simply provide a solution that is always an integrated approximation formed from all—in as much as one can feasibly obtain and curate relative to context—that has been measured before.

In any case, what emerges from these considerations is a systems tool, that offers clear and relatively simple equations/encodings that are usually quite easy to intuit, and straight forward to analyze using ones basic calculus background. These formulations are capable of revealing potentially deep underlying structure littered throughout real world data sets. To provide a simple framework to start to see the operation of this approach, consider the BCI applications above. In this context the ability of SAFIA to develop analytic neuroscientific, psychological, and sociological tools—for a relatable example—becomes immediately apparent.

Consider a subject  $S_i$  with  $J_\ell$  repeated experimental instances, each forming solution triples  $\{\mathbf{x}, \mathbf{c}, \mathbf{f}\}_\ell$  for  $\ell = 1, \dots, K$ , where  $\ell(J_\ell)$  indexes a stochastic solution from each experiment observed  $J_\ell$  times. Let the resulting inferred distributions transformed through  $\mathbf{c}$  by  $\mathbf{f}$  be  $d_\ell$ , each representing some  $\ell$ th trait of subject  $S_i$ . Now consider another stochastic trait, call it  $e_l$ , coming from either a different subject  $S_{i\pm 1}$  or being a different trait of subject  $S_i$ , i.e. for some index,  $e_k \equiv d_l$  if  $S_{i\pm 1} \equiv S_i$ . In either case, now consider the set of grouped traits  $\{d_\ell, e_l, \dots\}$  of all subjects of interest, and index it as the  $g$ th trait family  $\mathcal{T}_g$ .

Let  $\mathcal{T}_g$  be one of  $L$  trait families forming a meta-trait family  $\mathfrak{T}_L = \{T_1, \dots, T_g, \dots, T_L\}$  denoting collections of trait families grouped among some subset of subjects. We can now recurse the process from above, by taking triples of these grouped trait families and constructing shared distributions among them, call them

$\mathcal{D}_L$ , which then can also be easily represented by its own triple  $\{X, A, F\}_{\mathcal{D}_L}$ , which is now a low dimensional representation of the resulting distribution, unconditionally stable, and extremely cheap to analyze.

The meta-trait families  $\mathfrak{T}_L$  can similarly be recursed to form increasingly more general collections of trait groups, which can be used to examine increasingly more complex epiphenomena. These nested groupings of epiphenomena, similarly, have enveloping categories of increasing higher order epiphenomena, all of which can be containerized inside of simple mathematical representations, and examined in their analytic interdependencies and functional, as well as functorial, relationships. Of course, as the data is enriched as a function of time, transfer learning can be readily employed, to constantly update and adapt the representational models. This may indeed be necessary, because of seemingly curious features of consciousness [**<empty citation>**], which imply temporally nonlocal traits even at the biological level. This opens up a new temporal analysis feature ... (facebook Kats library?)

The goal of a SAFIA is different than many more conventional methods. SAFIA does not try to funnel every observed sub-phenomena into an abstract ideal, e.g. some mathematically static high dimensional model that is constrained to incredibly complex and computationally expensive representations, nor to bury the reader in easily constructed categorical formalism; but rather, to develop and simplify measured observations into intuitively penetrable representations, so as to reveal their most primitive dynamical features, relations, and ramifications. The details of this process can be made as precise as needed for any particular application, but the point is that it is ultimately up to the user to determine these choices. As such, SAFIA is a method designed to extract, quantify, predict, and analyze advanced integrated epiphenomena, using a tool chain that is shockingly tractable.

To make this clearer, we focus on the ASCERTAIN data set [Sub+16] for a demonstration of the method.

### 3.2. The ASCERTAIN data set example

The ASCERTAIN data set is a multimodal set of data aimed at identifying personality traits alongside the emotional states that arise while watching evocative movie scenes. In the study, an individual’s personality traits were identified along five dimensions using the common *big-five personality traits*, acquired from big-five marker scale questionnaires that participants were required to complete [PD02]. The five personality dimensions utilized along this scale are thought to correspond with: Extraversion (sociable versus reserved), Neuroticism (nervous versus confident), Agreeableness (compassionate versus dispassionate), Conscientiousness (dutiful versus easy-going) and Openness (curious versus cautious).

Alongside the personality tests, each participant was seated in front of a computer monitor, and fitted with GSR electrodes on the left index and middle fingers, facial landmark features recorded via webcam (EMO [Joh+11]), ECG sensors at each arm crook with a reference electrode on the left foot, and a dry-electrode EEG headset. All data was recorded using commercially available and minimally invasive sensors, allowing for free body movements. Each participant then viewed 36 movie clips between 51-127 seconds long, over a period of 90 minutes. After viewing each movie clip, the participant self-reported on an arousal-valence (AV) plane. In addition self-reported rating scales for engagement (from inattentive to highly attentive), familiarity (from never seen it, to know it well), and liking (from hated it to loved it) were recorded. For more details on the experimental design please refer to [Sub+16].

For this study we use the raw signals

#### 3.2.1. Data fusion

#### 3.2.2. Isolating meta-traits along multivariate contours

#### 3.2.3. Extracting meta-traits via cluster analysis

#### 3.2.4. Identifying signal distributions inferred from meta-traits

##### 3.2.4.1. Treating signal distributions independently.

##### 3.2.4.2. Merging signal distributions into meta-signals.

3.2.5. *Clustering meta-signals and meta-traits to extract novel epiphenomena*

3.3. *Automation towards the establishment of transpersonal ontologies*

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